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December 8, 2025

Professor Staci Hepler
Department of Statistical Sciences
Wake Forest University

Dear Professor Staci Hepler and Search Committee,

After completing my PhD in musicology in 2016, I was fortunate to become an Adjunct Assistant Professor at Wake Forest University (WFU) and taught MUS 101, Introduction to Western Music, in Fall 2017. The experience of designing and teaching a course at such a vibrant liberal arts college was engaging and joyful. I loved teaching on a campus where education for undergraduate students is the focus and students are viewed as colleagues. Without full-time employment opportunities in musicology, I went back to school and pursued a master's degree in statistics and then a PhD. Now nearing the end of my PhD in statistics, I am beyond excited to see two openings for Teacher-Scholar Postdoctoral Fellow at WFU. I am ready to again teach at a liberal arts college such as WFU and hope to collaborate with Leo Cella, with whom we share the same PhD advisor—Dr. Ryan Martin at NC State.

During my PhD study in statistics, I take full advantage of the wide range of faculty expertise offered by my department at NC State University and work on three research projects in different directions broadly connected by the concept centering uncertainty quantification (UQ). My first project, with Dr. Brian Reich, is about spatial and causal inference; we developed a fully Bayesian, spectral method to account for unmeasured spatial confounding and provide unbiased effect estimates and conservative credible intervals for multiple exposures and multiple outcomes. Originating from a summer internship at the Lawrence Livermore National Laboratory, my second project concerns surrogate modeling for computer experiments. With Dr. Kevin Quinlan (LLNL) and Dr. Annie Booth (Virginia Tech), I developed an active learning method that utilizes Gaussian processes as surrogate models and joint predictive probabilities as acquisition functions to efficiently find sampling points to locate the optimal design for contour locations from multiple computer experiments. After two projects motivated by applications, I turned to fundamentals and am currently applying possibilistic inferential models (IM) to decision theory with Dr. Ryan Martin and Dr. Jonathan Williams. Using possibility measures, IM incorporates imprecision and provides validity guarantees. These projects provide breadth in thinking and wide opportunities for current and future collaboration. I am especially keen on continuing the research on my current project—possibilistic IM—and would love to collaborate with Leo Cella, who has been collaborating with Dr. Martin since his time at NC State.

My teaching experience in statistics thus far includes being a teaching assistant in two master's level statistics courses (ST 501 & ST 503—Fundamentals of Statistical Inference I & Fundamentals of Linear Models and Regression) and two guest lectures for upper-level undergraduate statistics courses (Introduction to Mathematical Statistics I) in Fall 2025. Being a teaching assistant to master's students, I maintained a friendly environment in office hours. In the classroom, I provided motivation for the problems (e.g. Jacobian transformation), demonstrated the different methods for transformation for random variables, and checked in with students regularly to see if there were questions. In Summer 2025, I mentored six undergraduate students from NC State and two high school students from NC School of Science and Mathematics on data science projects. The one-on-one advising experience provided me with a sneak peek of how to mentor junior researchers. My teaching and advising experience helped me form the teaching philosophy of emphasizing critical thinking and the ability to learn over learning details. I am eager to work with undergraduate and graduate students at WFU, guiding them in learning statistics as non-majors as well as conducting statistical research as majors.

Having lived in Winston-Salem for 10 years, every visit feels like homecoming. If given the opportunity to work at WFU, I hope to have Leo Cella as my mentor and collaborate with him on research about possibilistic inferential models. I also plan to actively seek collaboration opportunities with other departments within WFU and maintain collaborations with external institutions such as LLNL. Thank you for your time and consideration.

Sincerely,

Shih-Ni Prim

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